

Luis A. Rodriguez-Condit

Technical Architect – Embedded & Web Systems

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Summary

Software architect and developer with over 20 years of software development experience, specializing in bringing complex, low-level systems to modern web applications that have been used by millions. My languages of choice are JavaScript, TypeScript, Java, and C/C++. I thrive owning solutions from end to end, a proven technical leader, skilled in translating business needs into user and technical requirements, driving code quality by advocating for clean and maintainable practices, and mentoring teams to deliver robust, performant, and user-centric solutions.

Skills

Languages	C, JavaScript, TypeScript, CSS, HTML, Java, C++, Python, Objective-C, Swift, z80 Assembly, Brew, J2ME
System and low level	Emscripten, Markdown, WebAssembly, Java Native Interface, Hardware Emulation, WebSerial, WebUSB, USB Communication, Cross-Compilation
Mobile/Web/Frameworks	Angular, Spring Boot, Lua, Gulp, Ant, Mocha, Jasmine, Selenium WebDriver
Leadership	Test-Driven Development, Technical Leadership, Agile/Scrum, Project Coordination, Git

Transferrable Skills

Technical Architecture	Full-Stack Development, System Integration, Low-Level Bridging, Cross-Compilation, Architected and implemented solutions that bridge proprietary embedded systems (C/z80) with modern web applications (Angular/TypeScript) via WebAssembly and Emscripten.
Web/Front-End Expertise	Angular, TypeScript, Responsive Design, API Integration (RESTful), Designed and implemented Angular applications for calculator emulation, license activation, and user utilities, utilizing third-party SSO and communicating with Java/SpringBoot middleware.
Embedded & Systems	Hardware Emulation, Low-Level Programming (C, z80), Communication Protocols (USB, WebSerial/WebUSB, JNI), Designed the glue layer for targeting embedded code on the web. Established cross-platform USB communication architecture using abstraction layers and JNI.
DevOps & Automation	Build System Optimization, CI/CD, Scripting (Gulp, Node.js, Jenkins), Deployment, Developed and maintained complex build scripts, optimizing workflow into parallel Jenkins jobs, resulting in a 70% build time speed-up.
Technical Leadership	Scrum Master, Technical Requirements, Cross-Functional Coordination, Code Review, Led development teams, serving as Scrum Master and Development Lead. Authored technical requirements based on user needs. Coordinated business-wide tech sharing sessions.

Mentorship & Team Morale	Empathetic Leadership, Mentoring, Knowledge Transfer, Managed and mentored engineering teams, offering tutorials and code deep-dives to improve team proficiency and maintain morale.
Software Quality & Testing	Test-Driven Development (TDD), Automation Frameworks, Code Quality Metrics, Prototyped automation frameworks (Node.js, Mocha, Selenium WebDriver). Designed and met code quality standards (unit testing, static analyzers, mandatory code reviews).
Product & User Focus	User-Centered Design, Accessibility Implementation, Requirements Analysis, Prioritized user experiences for simpler, quicker solutions. Implemented code in z80 and JavaScript to enable screen reader accessibility for calculator emulators.
Risk & Compliance	Open Source Auditing, Security Remediation, Responsible for reviewing third-party library lists to ensure eligibility and manage licensing. Led teams to address IT security scan concerns (modifying HTTP headers).

Work Experience

Texas Instruments -- Software Architect

Tenure involved an evolution from rapid functional prototyping (building POCs and starter code, thus, reducing time-to-market) and deep systems engineering to technical leadership roles and architectural oversight. Considered the go-to person for bringing low-level hardware emulation and proprietary communication protocols to desktop and web applications. As part of the metrics task force, designed and met code quality metrics with tools like unit tests, code sanitizers, linters and static analyzers. This career progression was formally recognized with an appointment to the first rung of TI's Technical Ladder.

Open Source Auditing

2018-2025

Responsible for reviewing software third party library lists to determine their eligibility for integrating into our products. End result of this per-release activity is a list of attributions that would get appended to the EULA for each shipped product.

Tech Sync Coordinator

2018-2025

Coordinated business-wide info sharing sessions and recruited presenters for the same. This weekly series resulted in broader understanding of deep technical topics as well as business-related subjects

Calculator Web Apps

2025

Led team that implemented the functionality for 2 web apps: One for USB connectivity with the calculator and another shell for the emulator of the calculator. Serving functions of development lead, scrum master. Reviewed user requirements and wrote technical requirements. Web apps were written in Angular, with libraries compiled from C via Emscripten. For USB connectivity, used WebSerial API.

Calculator File Transfer Library

2024-2025

Designed and implemented functionality for performing file transfers between 1) Web app and physical calculator using WebSerial API and 2) A webapp that contains an emulated calculator, making the API between both libraries match so they can be a drop-in replacement for each other

Calculator Transformation Graphing

2024-2025

Implemented functionality to support animating graphs on new hardware platform written in C, using the APIs provided by the calculator's tools.

Calculator Build Environment

2020-2025

Developed and maintained gulp.js scripts that automated the complex process of invoking disparate platform build tools, such as CMake and MSBuild. This build system was later optimized for speed by restructuring the workflow into parallel Jenkins jobs, resulting in a 70% build time speed-up. The task division was actively managed and rebalanced to sustain peak performance.

Calculator Device Emulation

2020-2025

Designed and implemented the glue layer for targeting embedded code on the web by transpiling to WebAssembly. Coordinated cross-functional team that did development and test for new features. Designed and implemented an automation entry point using WebSockets.

TI-84 Plus CE Online Calculator

2022

Implemented Angular application that provides access to an emulator for the TI 84 Plus CE calculator on web browsers. Implemented single-sign-on usage for user authentication (OAuth identity) and license validation (entitlement) via 3rd party provider. This was the first publicly available TI calculator on a web app with *over 1500 users in first month.

License Activation Center

2020-2021

Designed and implemented a key validation interface in coordination with our DBA to provide users a means to activate license codes. Front end implementation written in Angular with a RESTful middleware implemented in Java+SpringBoot communicating with an Oracle database.

TI-84 Plus CE Chrome App

2020

In reaction to the covid pandemic, led the team that worked to support remote learning on Chrome OS allowing students coming back from spring break to have a TI calculator to work with. To date, the application has been used by over 3.5 million users. -- written in angular reusing previous work from the ExamCalc product.

Automated Test Framework Prototype

2019

Prototyped automation framework for interacting with calculator libraries in the browser using Node.js, CucumberJS, Mocha and Selenium Web Driver.

Web Apps Server-Side

2019

Led team in determining Tomcat and Apache Web server configuration changes needed to address concerns brought up by IT security scans. Changes included modifying HTTP headers and additions to meta tags in the web app's HTML

TI-Nspire Connect

2019

Designed and implemented Angular shell to run the web app, as well as the JavaScript and C glue layers for cross-compiling NavNet, the TI-Nspire line of calculator's communication library with Emscripten to enable communicating with a TI-Nspire CX II device via WebUSB. 28k on it's first year.

ExamCalc Accessibility

2017-2018

Implemented code needed in both z80 assembly and javascript to support market needs to make calculator emulators interact correctly with a screen reader for the vision impaired. Due to the nature of the calculator emulators, no _aria_ element is readily available, so the emulator needs to provide the messages to be fed to a screen reader.

TI Connect For Chrome OS

2015-2016

Ported the communications layer on a utility that talks to TI calculators using Chrome's USB API. Implemented user-facing features in AngularJS in combination with Chrome APIs.

TI Connect CE

2013-2015

Delivered cross-platform compatibility for a proprietary CARS communication protocol by porting the codebase to generate platform-native libraries (Windows/macOS). Established the USB communication architecture using an abstraction layer and integrated the functionality into the application via Java Native Interface (JNI).

TI SmartView (Hardware Emulation)

2013-2016

Wrote and reviewed user requirements. Designed and implemented features related to interacting with emulated hardware. Helped the development team solve memory and performance issues.

TI 84 Plus CE ExamCalc

2013

Implemented emulators for ez80-based devices which have gone through our client's pilots involving 800k users without any reported issues.

ExamCalcs

2012-2013

TI calculators that are approved for use on *ACT®, SAT®, and AP® exams*, as well as most *state assessments.* . Ported and implemented calculator emulators to the web. Adapted z80 opcodes and emulated the TI-84 Silver Edition and TI-84 Plus C's ASICs. Ported Toshiba T4 cpu emulation from Java and implemented the TI-108 and TI-30MV's ASICs. Pilots involving 800k users without any reported issues.

<https://education.ti.com/en/us/about/press-center/press-listing-content-module/6-20-2013-ccss>

Document Display Standardization

2012

Given the strong desire for documents to display with the same word-wrapping and character counts on all supported devices (iPad being a new addition): researched and developed guidelines and layout rules for document rendering when being displayed on different screen resolutions.

Lua Glue

2011

Designed and implemented the infrastructure for creating new classroom activities using the Lua scripting language. The C counterpart for the Lua bindings, as well as the Lua engine itself gets cross-compiled to run on different platform including iOS, Nucleus and desktop operating systems.

Image Question type

Designed and implemented some of the infrastructure necessary to create new classroom activities using the Lua scripting language. The C counterpart for the Lua bindings, as well as the Lua engine itself gets cross-compiled to run on different platform including iOS, Nucleus and desktop operating systems. The use of Lua would open up the possibility of expanding TI-Nspire's question types in the future without needing to write them in Java

Ui And Background Thread Refactor

Worked in conjunction with the development team to refactor the usage of JNI, improving performance and memory usage.

Digital Chocolate -- Game Engineering Manager

Managed the team doing porting for the US carriers' devices.

Game Engineering Manager

2009-2010

Coordinating project planning for various carriers in the US, working in conjunction with the sales and marketing teams to reduce production costs and improve revenue. Lead and mentor the engineering team, offering tutorials and code deep-dives where needed.

Brew In App Purchase Library

2010

Created client module for Qualcomm's Application Value Billing API (C/C++) in conjunction with the team in Bangalore, India. Allowing games to have in-app-purchases as well as promoting code reuse on future game titles

Star Invasion	2010
Supported team that created Alien Invaders by porting for small-screened and resource limited devices. Aided the creation team to reduce the footprint for their project; done in J2ME, targeting small-screened and resource limited devices.	
Rollercoaster Rush for iPhone	2010
Ported iPhone implementation of OpenGL ES game to Qualcomm's Brew framework	
LemonQuest -- iPhone Lead Programmer	
Developing games and applications for the iPhone OS. Coordinated development with a geographically distributed team. Prompted the company to implement updates to infrastructure to improve productivity.	
Tool Programmer	2008
Aided in the development of a stage editor for a game, done in Java.	
Circulate and Circulate Prologue	2008
Originally a PC based game; it was ported to the iPhone platform. Game used every user-friendly feature on the iPhone, including it's 3d sound capabilities with Open AL and it's accelerometer. Ported the DirectX implementation to the iPhone's OpenGL ES, converting 3d models from quads to triangles on the. Translated levels originally designed for desktop/mouse to be usable on a touch device. This game earned a bronze medal from pocketgamer.co.uk.	
iFit	2009
Wrote an application that provided pedometer functionality as well as fitness tips for the iPhone	
Gameloft -- iPhone Programmer	
Various roles, starting from Porting Engineer, Lead Developer, 3d Programmer and R&D Programmer	
Blockbreaker for iPhone	2007-2008
Developed and ported first prototype for the brick-breaker game using the Celestial framework for iPhone. The Mexico studio's team was the first to build a successful prototype which led to the studio being assigned more iPhone projects. Used C-implementation for OpenGL ES as graphics engine and sourced models from other platforms.	
Bikini Volleyball Mini Games	2007
Implemented mini-games that were included with the main volleyball game when built for devices that would support them	
3d Programmer	2006-2007
In charge of porting multiple game titles for resource-limited, but 3d-capable devices. All ports used Java's JSR184 3d implementation for mobile devices and were adapted by tweaking frustum setup and removing bottlenecks in graphics pipelines where possible, resulting in improved performance or, in some more extreme cases reducing models' polygon count to achieve acceptable frames-per-second and keep games playable	
Virgin Mobile Team Lead	2005-2006
Led team charged with porting games to Virgin Mobile's catalog of space-constrained devices, often needing to reduce games' scope and swap out graphics with lower-detailed versions in addition to performing micro-optimizations to improve performance	
Other Projects	
50 Chinese Words	2009
Wrote an application that reproduces the strokes necessary for writing chinese characters as well as playing their sound. Developed a Chinese character writing tutor for the iPhone.	
Big 2 Poker	2010

Wrote graphics engine using C and OpenGL ES. Created art for a card game for iPhone OS.

iAarti for Katha Films

2010

Wrote a religious-focused application that shows images, plays sounds and uses the iPhone's accelerometer to interact with the user.

Education

Centro de Investigación Científica y de Educación Superior de Ensenada (CICESE)

Completed coursework towards: Master's in Electronics and Telecommunications with specialization in control systems and automation, 2002-2005

Academic Project

Robot Control Over Ethernet

2004

Developed RT-Linux kernel modules to communicate a PC that performed the PD control algorithm and a driver PC that sent control signals and motor positions over TCP sockets with low-latency, minimal-jitter to control a 2-degree-of-freedom robot arm. Control PC had a Java Swing UI connected to the kernel module via Java Native Interface. The driver PC performed I/O operations directly to a data acquisition card from kernel space.

Instituto Tecnológico y de Estudios Superiores de Monterrey (ITESM)

Bachelor's in Electronics Systems Engineering, 1997-2001

Academic Project

Control And Automation Seminar

2021

Attended summer student exchange at the Université de Poitiers, France with a focus on control systems engineering.

Professional Development

- Learn React - [Codecademy](#) - 2025
- Bluetooth Low Energy Fundamentals - [Nordic Developer Academy](#) - 2025
- A Beginner's Guide to Linux Kernel Development (LFD103) - [Linux Foundation](#) - 2025
- Learn C++ - [Codecademy](#) - 2025
- iOS developer nanodegree - [Udacity](#) - 2015
- Front-end web developer nanodegree - [Udacity](#) - 2014-2015
- Website Performance Optimization - [Udacity](#) - 2014
- Object-Oriented JavaScript - [Udacity](#) - 2014
- Javascript Basics - [Udacity](#) - 2014
- Intro to HTML and CSS - [Udacity](#) - 2014
- HTML5 Game Development - [Udacity](#) - 2014
- jQuery - [Codecademy](#) - 2013
- Software Testing - [Udacity](#) - 2012
- Web Application Engineering - [Udacity](#) - 2012
- Artificial Intelligence for Robotics - [Udacity](#) - 2012
- Introduction to Artificial Intelligence - [Stanford](#) - 2011

DISCLAIMER: Portions of this document were created using an AI to tailor experience points to the target role. All content, including dates, titles, and technical claims, is 100% accurate and based on my verified career history. I have thoroughly proofread and confirmed its factual correctness.